In[10]:=
```
dist = PDF[NormalDistribution[xf + Deltaxv, sv], xhatv] ×
  PDF[NormalDistribution[xf, sp], xhatp] ×
  PDF[NormalDistribution[0, spert], Deltaxv] × PDF[NormalDistribution[b, sf], xf]
```

Out[10]=

$$\frac{e^{-\frac{\text{Deltaxv}^2}{2\,\text{spert}^2}-\frac{(-b+xf)^2}{2\,sf^2}-\frac{(-xf+xhatp)^2}{2\,sp^2}-\frac{(-\text{Deltaxv}-xf+xhatv)^2}{2\,sv^2}}}{4\,\pi^2\,sf\,sp\,\text{spert}\,sv}$$

In[11]:=
```
Assuming[{sv > 0, sp > 0, spert > 0, sf > 0},
 Integrate[dist, {Deltaxv, -Infinity, Infinity}, {xf, -Infinity, Infinity}]]
```

Out[11]=

$$\frac{e^{-\frac{b^2\,(sp^2+\text{spert}^2+sv^2)+\text{spert}^2\,\text{xhatp}^2+sv^2\,\text{xhatp}^2+sf^2\,(\text{xhatp}-\text{xhatv})^2+sp^2\,\text{xhatv}^2-2\,b\,(\text{spert}^2\,\text{xhatp}+sv^2\,\text{xhatp}+sp^2\,\text{xhatv})}{2\,(sp^2\,(\text{spert}^2+sv^2)+sf^2\,(sp^2+\text{spert}^2+sv^2))}}}{2\,\pi\,sf\,sp\,sv\,\sqrt{\frac{1+\frac{\text{spert}^2}{sv^2}}{sf^2}+\frac{sp^2+\text{spert}^2+sv^2}{sp^2\,sv^2}}}$$

In[12]:=
```
Simplify[%11]
```

Out[12]=

$$\frac{e^{-\frac{b^2\,(sp^2+\text{spert}^2+sv^2)+\text{spert}^2\,\text{xhatp}^2+sv^2\,\text{xhatp}^2+sf^2\,(\text{xhatp}-\text{xhatv})^2+sp^2\,\text{xhatv}^2-2\,b\,(\text{spert}^2\,\text{xhatp}+sv^2\,\text{xhatp}+sp^2\,\text{xhatv})}{2\,(sp^2\,(\text{spert}^2+sv^2)+sf^2\,(sp^2+\text{spert}^2+sv^2))}}}{2\,\pi\,sf\,sp\,sv\,\sqrt{\frac{1+\frac{\text{spert}^2}{sv^2}}{sf^2}+\frac{sp^2+\text{spert}^2+sv^2}{sp^2\,sv^2}}}$$

In[14]:=
```
unpert = PDF[NormalDistribution[xf, sv], xhatv] ×
  PDF[NormalDistribution[xf, sp], xhatp] × PDF[NormalDistribution[b, sf], xf]
```

Out[14]=

$$\frac{e^{-\frac{(-b+xf)^2}{2\,sf^2}-\frac{(-xf+xhatp)^2}{2\,sp^2}-\frac{(-xf+xhatv)^2}{2\,sv^2}}}{2\,\sqrt{2}\,\pi^{3/2}\,sf\,sp\,sv}$$

In[15]:=
```
Assuming[{sv > 0, sp > 0, sf > 0}, Integrate[unpert, {xf, -Infinity, Infinity}]]
```

Out[15]=

$$\frac{e^{-\frac{b^2\,(sp^2+sv^2)+sv^2\,\text{xhatp}^2+sf^2\,(\text{xhatp}-\text{xhatv})^2+sp^2\,\text{xhatv}^2-2\,b\,(sv^2\,\text{xhatp}+sp^2\,\text{xhatv})}{2\,(sp^2\,sv^2+sf^2\,(sp^2+sv^2))}}}{2\,\pi\,sf\,sp\,\sqrt{\frac{1}{sf^2}+\frac{1}{sp^2}+\frac{1}{sv^2}}\,sv}$$